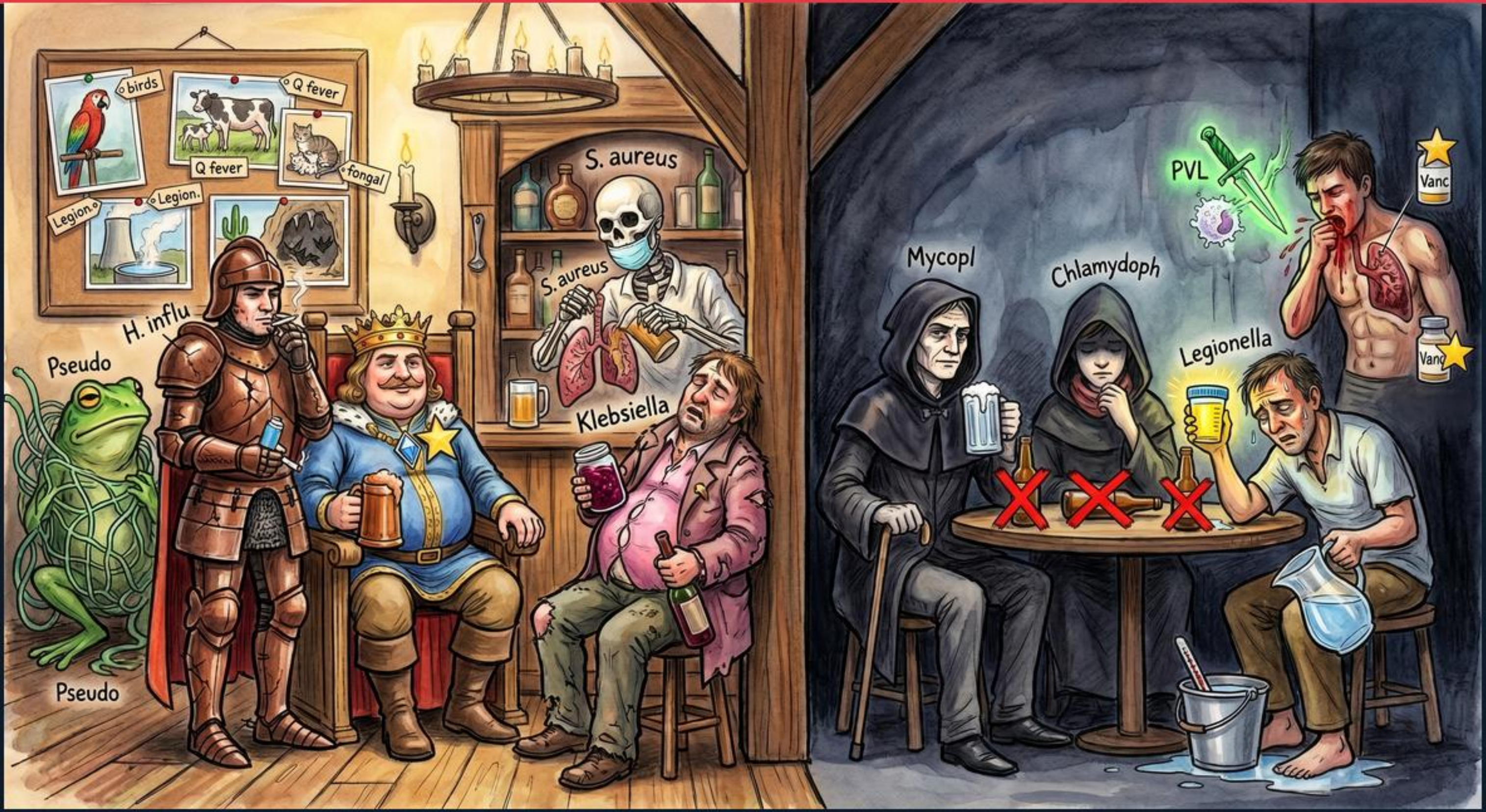
Pneumonia — Organisms (Typical, Atypical, Epidemiologic Clues & CA-MRSA)



13 symbols · sketch walkthrough

S. pneumoniae (crowned king)

WHAT YOU SEE

Fat crowned king with a gold star clutching a beer mug — the #1 reigning bacterial cause of CAP.

WHAT IT ENCODES

S. pneumoniae is the #1 bacterial cause of CAP: a lancet-shaped, alpha-hemolytic, optochin-sensitive, bile-soluble gram-positive diplococcus with a polysaccharide capsule (main virulence factor and vaccine target). Classic lobar pneumonia with rust-colored sputum; most common cause of bacteremic pneumonia. Resistance is via altered penicillin-binding proteins (PBP), NOT beta-lactamase — overcome with high-dose amoxicillin 1 g TID.

H. influenzae (knight)

WHAT YOU SEE

Armored knight drinking beside the king — a 'typical' that gram-stains despite the misleading name.

WHAT IT ENCODES

H. influenzae is a TYPICAL gram-negative coccobacillus that gram-stains and grows on chocolate agar (needs factors X/hemin and V/NAD). Leading CAP organism in COPD/smokers and the elderly. Empiric beta-lactams cover it, but nontypeable strains may produce beta-lactamase — use amox-clavulanate, not plain amoxicillin.

THE BOARD TRAP

"atypical because the name sounds like flu" -> wrong; H. influenzae IS a typical — it gram-stains and cultures on enriched media, unlike Mycoplasma/Chlamydia/Legionella.

Klebsiella (pink drunk)

WHAT YOU SEE

Disheveled drunk in a pink shirt slumped with a bottle — encapsulated Klebsiella in the alcoholic.

WHAT IT ENCODES

Klebsiella is a heavily encapsulated (mucoid) gram-negative rod causing severe lobar pneumonia in alcoholics, diabetics, and aspiration. Buzzwords: 'currant-jelly' sputum, bulging fissure sign on CXR, early cavitation/necrosis with high mortality. Treat per susceptibilities (3rd-gen cephalosporin, or carbapenem if ESBL).

THE BOARD TRAP

"uncomplicated pneumococcal lobar pneumonia" for a cavitating alcoholic -> wrong; cavitation/abscess in an alcoholic is Klebsiella or anaerobes; pneumococcus rarely cavitates.

S. aureus (skeleton)

WHAT YOU SEE

Masked skeleton pouring beer behind the bar — necrotizing post-influenza S. aureus.

WHAT IT ENCODES

S. aureus causes necrotizing, often cavitary pneumonia, classically as a post-influenza superinfection and in IV drug users (septic emboli). Clues: pneumatoceles, multilobar infiltrates, rapid cavitation. MSSA = nafcillin/cefazolin; MRSA = vancomycin or linezolid.

THE BOARD TRAP

"viral relapse" for post-flu worsening with new cavities -> wrong; that is bacterial superinfection (S. aureus/pneumococcus); ceftriaxone does NOT cover MRSA.

Pseudomonas (green frog)

WHAT YOU SEE

Green frog crouched at far lower-left — Pseudomonas in the structurally damaged lung.

WHAT IT ENCODES

Pseudomonas is an oxidase-positive, non-lactose-fermenting gram-negative rod infecting structural lung disease (bronchiectasis, CF, severe COPD), prior broad-spectrum antibiotics, frequent steroids, or recent hospitalization. Cover with an antipseudomonal beta-lactam (pip-tazo, cefepime, meropenem); add a 2nd antipseudomonal in severe disease until susceptibilities return.

THE BOARD TRAP

"add antipseudomonal coverage to all CAP" -> wrong; reserve it for documented risk factors (structural lung disease, recent abx/hospitalization, prior Pseudomonas).

Epidemiologic clue wall

WHAT YOU SEE

Framed picture gallery — parrot, cows, cat, Q-fever and Legion placards — exposure history fingerprints the bug.

WHAT IT ENCODES

Exposure history pins the pathogen: hotel/cruise/water -> Legionella; birds/parrots -> Chlamydia psittaci; farm animals/parturient cattle or cats -> Coxiella burnetii (Q fever); Southwest deserts -> Coccidioides; Ohio/Mississippi caves/bats -> Histoplasma; rabbits -> Francisella (tularemia). Always ask travel, animals, occupation.

Mycoplasma (cloaked, X'd)

WHAT YOU SEE

Cloaked figure with a beer, red X on the table — Mycoplasma 'walking pneumonia,' beta-lactam resistant.

WHAT IT ENCODES

Mycoplasma is classic 'walking pneumonia' of young healthy adults (teens, military, dorms): mild interstitial infiltrates out of proportion to a well-appearing patient. NO cell wall (cannot gram-stain; beta-lactams useless). Extrapulmonary clues: cold-agglutinin hemolytic anemia, bullous myringitis, erythema multiforme/SJS. Treat macrolide, doxycycline, or respiratory FQ.

THE BOARD TRAP

"beta-lactam" for Mycoplasma -> wrong; no cell wall means no beta-lactam target — use macrolide/doxycycline/respiratory FQ.

Chlamydophila (cloaked, X'd)

WHAT YOU SEE

Second hooded figure, red X — Chlamydophila pneumoniae, an obligate intracellular atypical.

WHAT IT ENCODES

Chlamydophila pneumoniae is an obligate intracellular atypical causing mild, often biphasic CAP (pharyngitis/hoarseness preceding cough) in young adults; <1% of CAP. Cell-wall-deficient and intrinsically beta-lactam resistant. Treat macrolide, doxycycline, or respiratory FQ.

THE BOARD TRAP

"beta-lactam will treat it" -> wrong; as an atypical it won't respond; distinguish from C. psittaci by bird/parrot exposure (more severe).

Legionella (cloaked, X'd)

WHAT YOU SEE

Third figure with beer and a bucket of water, red X — Legionella from aerosolized water.

WHAT IT ENCODES

Legionella pneumophila is a fastidious gram-negative atypical (needs buffered charcoal yeast extract agar) from aerosolized water — cooling towers, hot tubs, hotels/cruises, hospital plumbing. Severe CAP with high fever, relative bradycardia, diarrhea, hyponatremia, elevated LFTs, and confusion. Diagnose fast with urinary antigen (serogroup 1). Treat respiratory FQ or azithromycin.

THE BOARD TRAP

"negative urinary antigen excludes Legionella" -> wrong; it detects only serogroup 1 — culture/PCR if suspicion is high; never beta-lactam alone.

Three red X's = beta-lactam resistant

WHAT YOU SEE

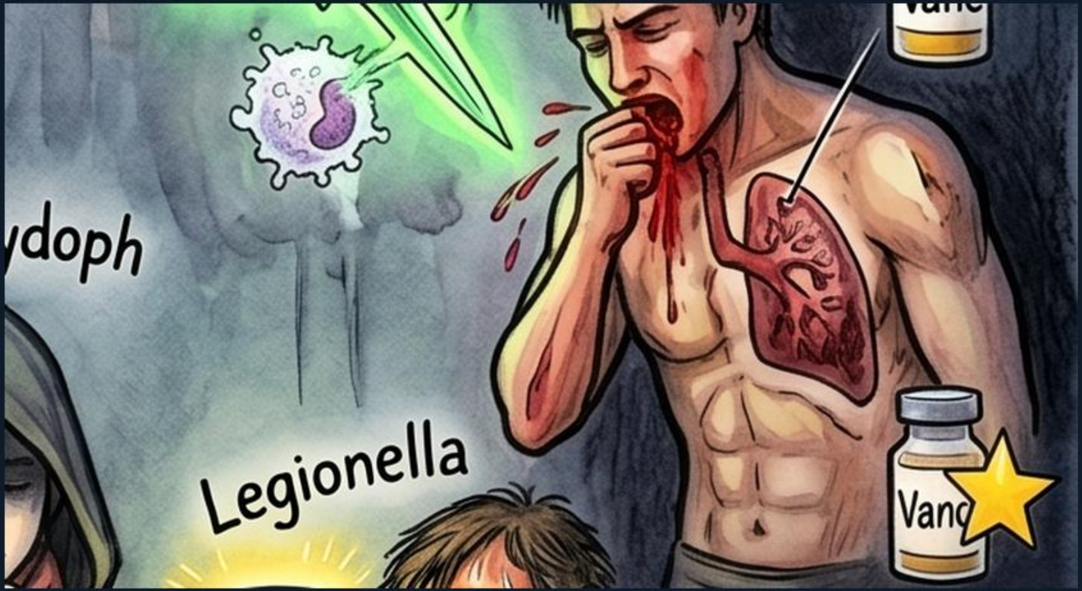
Row of three big red X's under the cloaked trio — atypicals are intrinsically resistant to all beta-lactams.

WHAT IT ENCODES

The atypicals (Mycoplasma, Chlamydophila, Legionella) are intrinsically resistant to ALL beta-lactams — no peptidoglycan target or intracellular location. They need cell-penetrating protein/DNA-synthesis agents: macrolides, tetracyclines, respiratory fluoroquinolones. This is exactly why empiric CAP pairs a beta-lactam WITH a macrolide (or uses a respiratory FQ alone).

THE BOARD TRAP

"failure on ceftriaxone = resistance" -> wrong; the real issue is an uncovered atypical — add macrolide/doxycycline or switch to respiratory FQ.



CA-MRSA hemoptysis man

WHAT YOU SEE

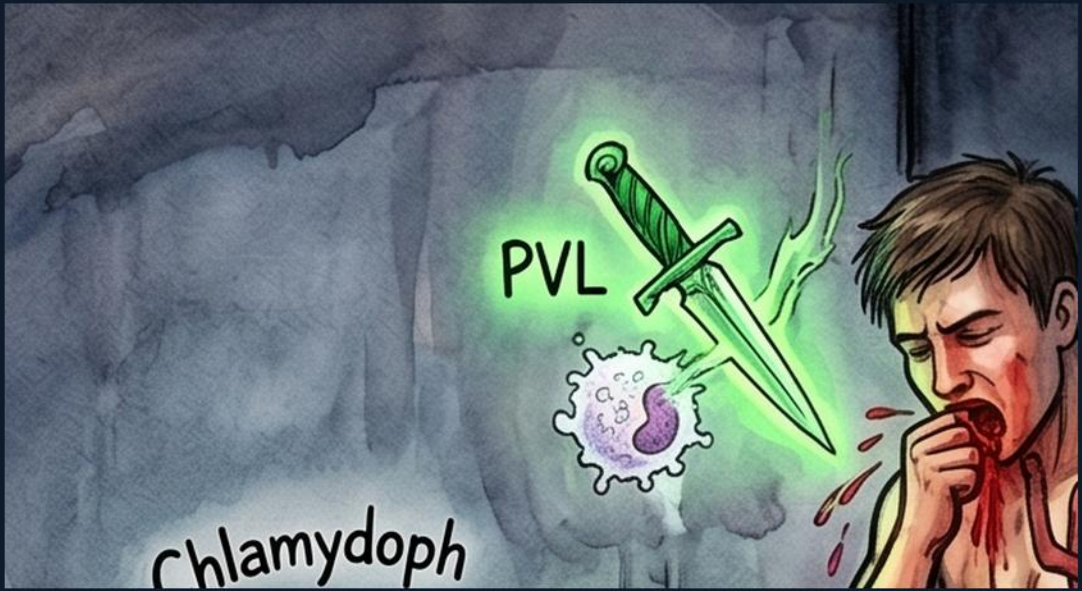
Shirtless man coughing blood holding a necrotic lung with a gold star — CA-MRSA necrotizing pneumonia.

WHAT IT ENCODES

CA-MRSA carries SCCmec type IV (mecA -> altered PBP2a) and often Panton-Valentine leukocidin (PVL), causing necrotizing, cavitary pneumonia with gross hemoptysis, profound leukopenia, and high mortality — often post-influenza in young healthy hosts. Severe disease: vancomycin or linezolid (linezolid also suppresses toxin).

THE BOARD TRAP

"ceftriaxone covers it" -> wrong; no beta-lactam covers MRSA except ceftaroline — escalate to vancomycin/linezolid.



PVL toxin (green dagger)

WHAT YOU SEE

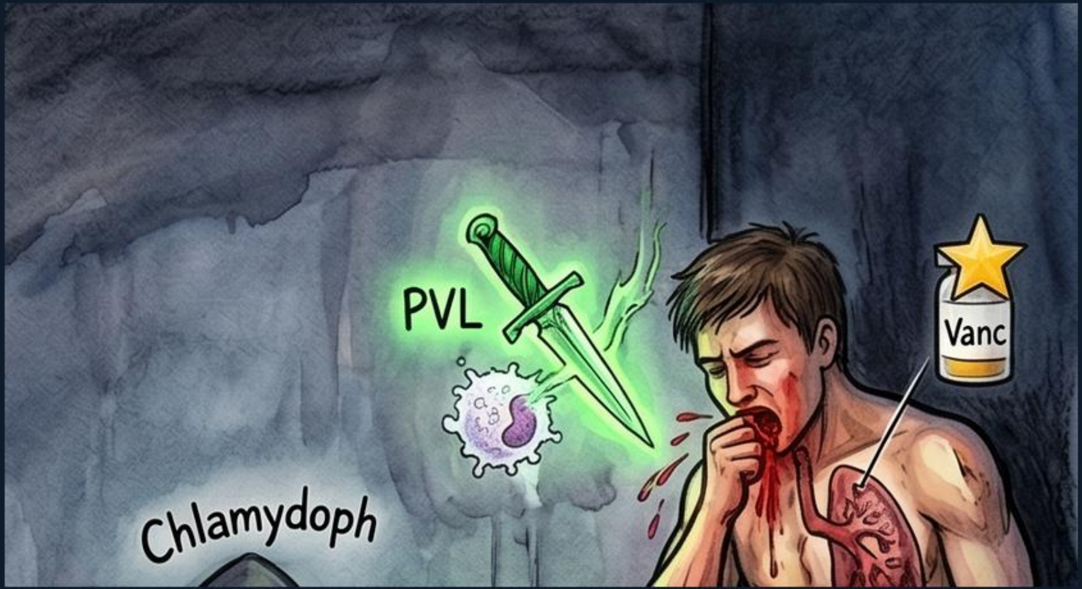
Glowing green dagger labeled PVL — the pore-forming leukocidin destroying neutrophils.

WHAT IT ENCODES

Panton-Valentine leukocidin (PVL) is a pore-forming exotoxin lysing neutrophils, driving the fulminant necrotizing, cavitary, hemoptysis-heavy pneumonia with leukopenia. Add a protein-synthesis inhibitor that SUPPRESSES toxin (linezolid, or clindamycin if susceptible) — vancomycin alone does not turn off toxin synthesis.

THE BOARD TRAP

"leukopenia is reassuring" -> wrong; in PVL necrotizing MRSA leukopenia is an ominous severity marker.



Vancomycin / linezolid (Vanc bottles)

WHAT YOU SEE

Two 'Vanc' bottles beside the bleeding man — the IV anchors for severe MRSA pneumonia.

WHAT IT ENCODES

Severe MRSA pneumonia is treated with vancomycin (AUC-guided) or linezolid; linezolid penetrates lung epithelial-lining fluid well and suppresses toxin. CA-MRSA is generally more susceptible than hospital MRSA — TMP-SMX, clindamycin, or doxycycline suffice for skin/soft tissue, but NOT for severe pneumonia. Check clindamycin D-test for inducible resistance.

THE BOARD TRAP

"oral TMP-SMX or doxycycline for severe MRSA pneumonia" -> wrong; those are for outpatient SSTI — use IV vancomycin/linezolid for severe lung disease.

Summary — every symbol

1. S. pneumoniae (crowned king)

S

2. H. influenzae (knight)

H

3. Klebsiella (pink drunk)

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5. Pseudomonas (green frog)

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7. Mycoplasma (cloaked, X'd)

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8. Chlamydophila (cloaked, X'd)

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10. Three red X's = beta-lactam resistant

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